SYSTEM DEVELOPMENT PLAN

SaleSync: Recommended System Plan and Database Design

1. Recommended features for each role

1.1 Customer

Essential features

- Register, log in, log out, and verify email.
- Can view only products if not signed in.
- Browse, search, filter, and view product details.
- View seller/store information.
- Add products to a cart and change quantities.
- Checkout with an order summary.
- Pay through PayMongo Test Mode.
- View order history, payment status, and order status.
- Manage basic profile information and delivery address.
- Receive essential order and payment notifications.
- Multiple advanced address-book options.

Remove or defer

- Product comparison.
- Wishlist and saved products.
- Advanced recommendations and personalized feeds.
- Complex support conversations.
- Product reviews and ratings.
- Return/refund request management, unless specifically required.

Recommended customer navigation: Home, Shop, Cart, My Orders, and Profile.

Keep the navigation short. Avoid creating a separate page for every minor action when a section or modal can handle it clearly

1.2 Seller

Essential features

- Apply to become a seller (Seller Application Module)
- View application status and store approval.
- View a simple store dashboard.
- Add, edit, publish, and archive products.
- Manage product price, description, SKU, and stock quantity.
- View orders containing their products.
- Update permitted order fulfilment statuses.
- View a basic sales summary for their own store.

Remove or defer

- Advanced analytics and forecasting.
- Complex shipping-carrier integrations.
- Separate customer messaging systems.
- Advanced inventory movement dashboards.
- Multiple store customization options.
- Seller-specific review response management.
- Extensive sales exports, if the basic sales summary is enough for the project.

Recommended seller navigation: Dashboard, My Products, Orders, Sales Summary, and Store Profile.

1.3 Management Admin

Essential features

- Dashboard with customers, sellers, products, orders, and sales totals.
- View, activate, suspend, and restore customer accounts.
- Approve or reject seller applications.
- View and moderate product listings.
- View and oversee all orders.
- View payment statuses.
- Generate date-filtered sales reports.
- Export sales reports to CSV.
- View basic administrative activity history.

Remove or defer

- Technical server configuration.
- Database backup and restoration controls.
- Application-wide maintenance mode.
- Low-level email or payment API configuration.
- Advanced promotion campaigns and voucher rules.
- Complex customer support management, unless it is a specific requirement.
- Multiple overlapping dashboards that show the same statistics.

Recommended Management Admin navigation: Dashboard, Users, Seller Applications, Products, Orders, and Sales Reports.

1.4 Maintenance Admin

Essential features

- System health dashboard.
- Application settings for approved configuration values.
- Enable and disable maintenance mode.
- Create and list database backups.
- Restore a selected backup with confirmation.
- Review system configuration changes.
- Review security-relevant audit logs.
- View application, database, email, and payment integration status.

Remove or defer

- Product management tools that duplicate the Seller or Management Admin.
- Customer and seller account management.
- Sales and business analytics.
- A custom code editor.
- Arbitrary SQL execution from the browser.
- Complex deployment management.
- A full plugin marketplace or unrestricted sandbox environment.

Recommended Maintenance Admin navigation: System Overview, System Settings, Backup & Restore, Maintenance Mode, and Audit Logs.

2. Features that should be shared across all roles

These are system-wide functions. They should be implemented once in the backend and reused by the different roles wherever needed.

Shared feature	Implementation plan	Roles that use it
Authentication	Central login, logout, password hashing, sessions, and email verification	All four roles
Role-based access control	Backend middleware checks the user's role before granting access	All four roles
User and session management	Store user identity and account status centrally	All four roles, with different permissions
Product catalogue	One product database shared by customers, sellers, and Management Admin	Customer, Seller, Management Admin
Inventory	Central stock records and validated stock adjustments	Customer, Seller, Management Admin
Orders	Shared order creation, order items, status, and history	Customer, Seller, Management Admin
Payments	Central payment records and verified PayMongo Test Mode results	Customer, Management Admin

Notifications	Shared notification service for account and order events	All four roles, depending on event
Audit logs	Record important administrative, security, and maintenance actions	Management Admin and Maintenance Admin
Settings	Central settings service with role restrictions	Maintenance Admin; other roles read only approved public settings
Error handling	Standard error responses and server-side logging	All roles and backend services
Database backup	A centralized backup and recovery service	Maintenance Admin

2.1. Orders and payments

The backend should coordinate the entire purchasing process:

1. Validate the customer's cart.
2. Recalculate the subtotal and final amount.
3. Verify stock availability.
4. Create the order and order items.
5. Initiate the payment through PayMongo Test Mode.
6. Verify payment status using the provider's supported server-side mechanism.
7. Update the payment and order status.
8. Record the event and notify the relevant users.

Use database transactions for operations that must succeed together. Payment-provider calls need additional failure handling because a database transaction cannot make an external payment service atomic.

3. Recommended database design

The design below is a proposed simplified schema for your requirements.

3.1 Core tables

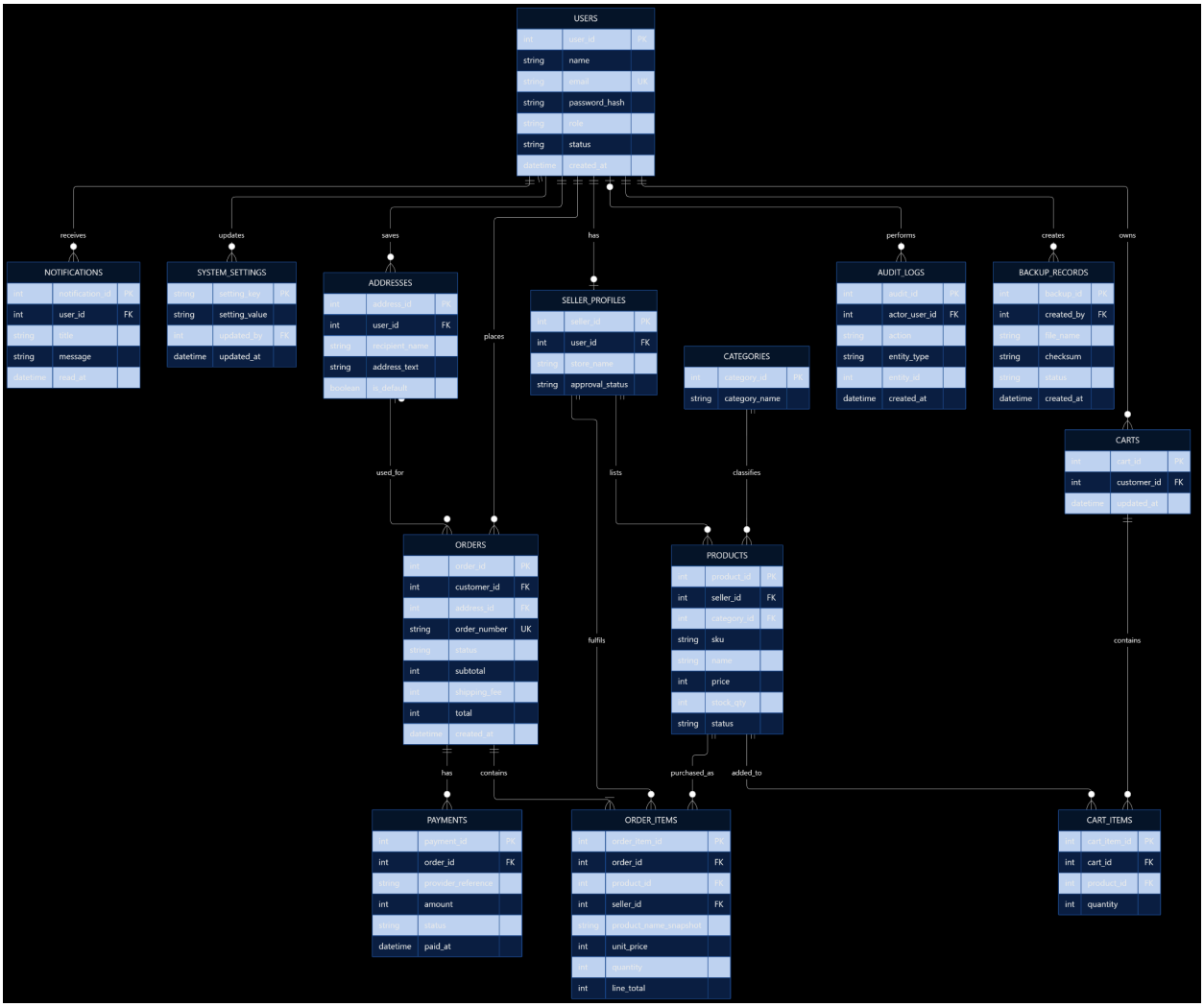
Table	Important fields	Purpose
`users`	`user_id` PK, `name`, `email` UNIQUE, `password_hash`, `role`, `status`, `created_at`	Central identity and authentication
`seller_profiles`	`seller_id` PK, `user_id` FK UNIQUE, `store_name`, `description`, `approval_status`	Seller application and store profile
`categories`	`category_id` PK, `category_name`	Product classification
`products`	`product_id` PK, `seller_id` FK, `category_id` FK, `sku`, `name`, `description`, `price`, `stock_qty`, `status`	Product listings and current stock
`addresses`	`address_id` PK, `user_id` FK, `recipient_name`, `address_text`, `is_default`	Customer delivery addresses
`carts`	`cart_id` PK, `customer_id` FK, `created_at`, `updated_at`	Customer shopping cart
`cart_items`	`cart_item_id` PK, `cart_id` FK, `product_id` FK, `quantity`	Products and quantities in a cart

orders	order_id PK, customer_id FK, address_id FK, order_number UNIQUE, status, subtotal, shipping_fee, total, created_at	Purchase transaction
order_items	order_item_id PK, order_id FK, product_id FK, seller_id FK, product_name_snapshot, unit_price, quantity, line_total	Individual items purchased
payments	payment_id PK, order_id FK, provider_reference, amount, status, paid_at	Payment attempts and results
notifications	notification_id PK, user_id FK, title, message, read_at, created_at	In-app notifications
audit_logs	audit_id PK, actor_user_id FK, action, entity_type, entity_id, created_at	Administrative activity history
system_settings	setting_key PK, setting_value, updated_by FK, updated_at	Approved configurable system values
backup_records	backup_id PK, created_by FK, file_name, checksum, status, created_at	Backup metadata and operation history

You may also add order_status_history to preserve every order status change and payment_events if you need a detailed record of PayMongo callbacks and payment state transitions.

3.2 Entity Relationship Diagram (ERD)

The following Mermaid ERD uses Crow's Foot notation.



4. How the four roles connect to the database

The database should not be divided into four separate databases. Each role accesses the same underlying records through different authorized operations.

Database entity	Customer	Seller	Management Admin	Maintenance Admin

Users	Read/update own permitted details	Read/update own permitted details	Manage customer and seller accounts	Manage own administrative account; technical recovery only when necessary
Seller profiles	Read public store information	Manage own profile and view application status	Approve/reject and manage all stores	No routine store management
Products	Read published products	Manage own products	Moderate all products	No routine product management
Categories	Read	Read	Manage categories if required	No routine category management
Cart and cart items	Manage own cart	No routine access	No routine access	No routine access
Addresses	Manage own addresses	No routine access	Restricted access only when needed for authorized operations	No routine access
Orders and order items	Read own orders	Read orders containing their items	View all orders and oversee business operations	No routine access

Payments	View own payment status	No access to sensitive payment data	View payment status and report totals	Check integration health without exposing secrets
Notifications	Read own notifications	Read own notifications	Read own administrative notifications	Read own maintenance notifications
Audit logs	No access	No access	Read relevant business administration events	Read system and maintenance events
System settings	Read approved public settings	Read approved public settings	No technical configuration access by default	Manage approved settings
Backup records	No access	No access	No routine access	Create and manage backups and restore workflows

5. Recommended system development plan

Recommended the following order of implementation. This minimizes duplicated work and helps your team avoid building features that later need to be rewritten.

01

Finalize scope and role permissions

Decide which features are essential, remove duplicate sections, agree on order/payment statuses, and document who can perform each action.

Deliverable: approved feature list and role-permission matrix.

02

Finalize the database and API contracts

Create the ERD, verify existing SQLite tables, define foreign keys and constraints, and agree on the backend endpoints before changing multiple modules.

Deliverable: approved schema, migrations, and API specification.

03

Complete authentication and authorization

Test all four roles and ensure unauthorized users cannot access protected routes or another user's data.

Deliverable: tested login and role restrictions.

04

Complete the main shopping workflow

Connect product browsing, cart, checkout, order creation, inventory updates, and payment status. Test the complete flow before adding optional features.

Deliverable: a customer can complete a test purchase and see the resulting order.

05

Complete seller and management workflows

Verify product ownership, seller approval, order fulfilment, account management, and sales reports.

Deliverable: sellers and Management Admin see accurate, permission-filtered data.

06

Complete maintenance and recovery

Finish the controlled settings sandbox, maintenance mode, backup creation, restore testing, and audit logs.

Deliverable: a backup can be restored and the operation is logged.

07

Add historical data and perform final testing

Generate at least 200 agreed dummy records spanning October 25, 2024, through October 25, 2026. Validate reports, role permissions, and recovery procedures.

Deliverable: repeatable demo data and a documented acceptance-test checklist.

6. Sales report rules to should establish

Before implementing the Management Admin's report panel, decide what each metric means. Otherwise, two pages might display different totals for the same date range.

Metric	Recommended rule
Total orders	Count orders created within the selected date range.
Paid orders	Count orders whose payments have been verified as successful.
Gross sales	Sum eligible paid order totals before refunds, according to your documented accounting rule.
Net sales	Gross sales minus recorded refunds, using a consistent definition.
Seller sales	Aggregate eligible sales associated with each seller's order items.
CSV export	Use the same date filters and calculations as the on-screen report.

IMPORTANT NOTES & FEATURES MENTIONED BY PROF. DALUYON:

- Move product creation sa System settings.
- Implement PayMongo.
- Add 200 dummy data (Past 2 years ata)
- Fix Sales Report like include lang need makita sa report pag inexport to CSV.

Seller Modules

Dashboard Module

- View total products.
- View total orders.
- View total sales.
- View recent order activities.

Product Management Module

- Add products.
- Edit product information.
- Publish products.
- Archive or deactivate products.
- Manage product price.
- Manage product description.
- Manage SKU.
- Manage stock quantity.

Order Management Module

- View orders containing their products.
- View order details.
- Update order status:
    - Processing
    - Ready for Shipment
    - Shipped
    - Completed

Sales Summary Module

- View total sales.
- View sales by date range.
- View number of orders.
- View best-selling products.

Store Settings Module

- View and update store profile.
- View store approval status.
- Manage basic store information (Store Name, Description, Contact Information).